



Benefits and Challenges of ICT & Computer Evolution

Grade 10 - ICT - Module 1.3

GRADE 10 · MODULE 1.3

Learning Objectives:

1. Identify benefits of ICT
2. Understand demerits/challenges of ICT
3. Explain computer evolution and generations
4. Recognize emerging ICT trends
5. Practice responsible ICT use

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Benefits of ICT - Overview

Key Point: "ICT has made man's life easy and convenient"

Benefits by Sector:

Sector	Benefits
Communication	Easy information exchange, social media, instant messaging
Education	Access to information, online learning, digital resources
Business	Global reach, automation, cost reduction, efficiency
Health	Advanced diagnostics, telemedicine, remote monitoring
Agriculture	Smart farming, automation, higher yields
Government	e-Government services, transparency, accessibility
Daily Life	Entertainment, convenience, time saving, connectivity

Benefits of ICT - Detailed

Detailed Benefits:

Benefit	Explanation
 Speed	Information processed and delivered quickly
 Global Reach	Connect with anyone anywhere
 Cost Reduction	Reduces travel, communication, and operational costs
 Efficiency	Automated tasks, 24/7 operations
 Access to Information	Knowledge at your fingertips
 Connectivity	Social media, video calls, collaboration

Real Examples:

"Today, man can process data using computers instead of pen and paper"

"Business communities can meet via video conferencing without travelling"

"Patients in remote villages can consult specialists via telemedicine"

Discussion Point:

"How has ICT made your daily life easier?"

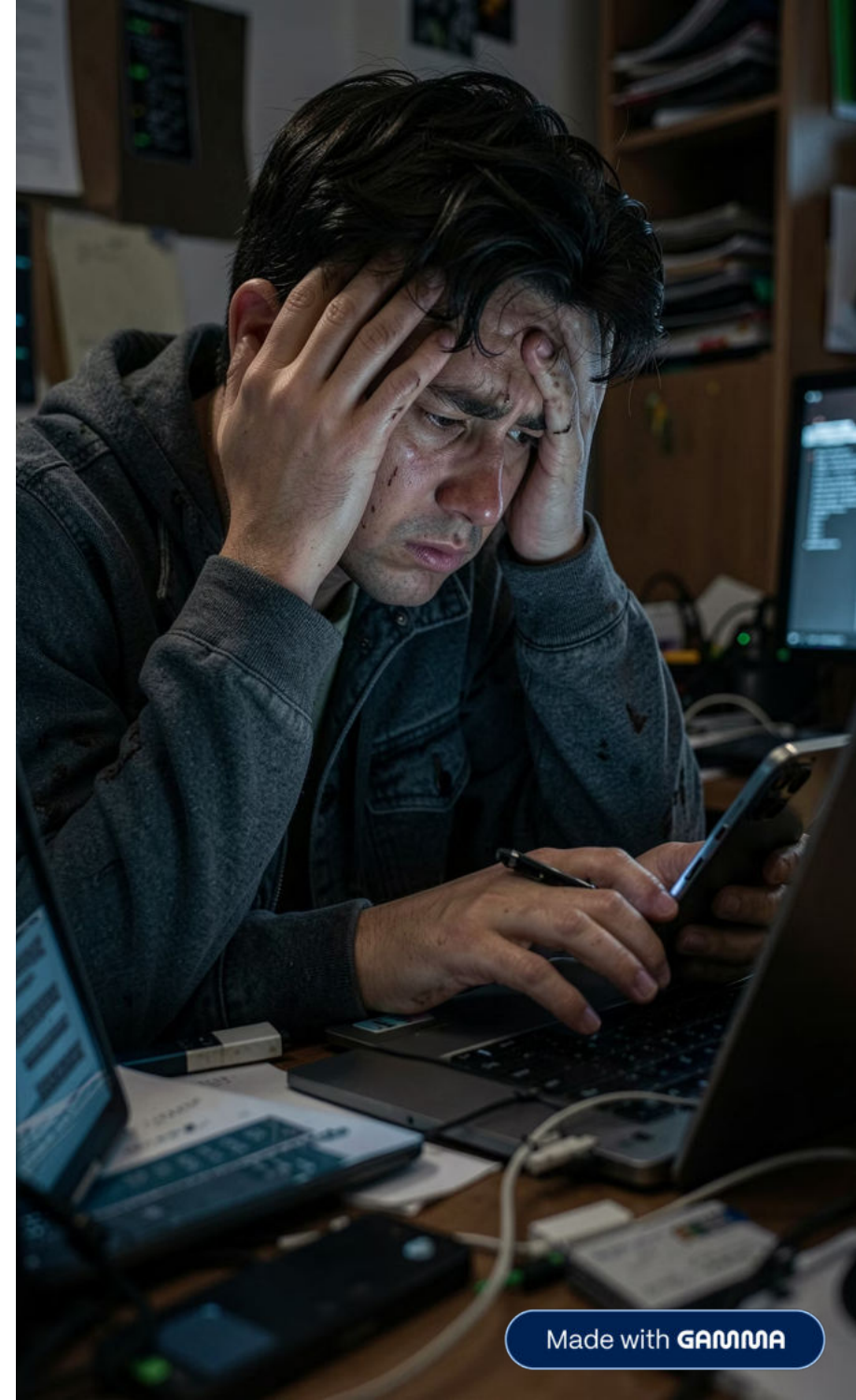
Demerits of ICT - Introduction

Important Note:

"Not only does ICT help us as an enabler to perform day-to-day activities, it has become a close companion. However, if you use this companion in an undue manner, it may bring forth undesirable effects."

Demerits of ICT:

- ● Addiction
- ● Health issues
- ● Inappropriate friendships
- ● Computer viruses
- ● Psychological effects
- ● Privacy violations



Demerits of ICT - Detailed

Detailed Demerits:

Addiction	Student/person immerses in computer/games without limit; may lose track of education
Health Ailments	Sore eyes, back pain, headache from excessive use
Inappropriate Friendships	Building unsuitable friendships through social media
Computer Viruses	Computers can become infected due to improper internet use
Psychological Effects	Negative effects on psychology by visiting improper websites
Privacy Issues	Edited photos/videos affecting personal lives

Key Point:

"There is an opinion that ICT has negative impacts on human society due to issues such as these. But we have understood that the systems made by ICT make most of man's routine work effortless."



Responsible ICT Use

"It is our responsibility to use ICT in a responsible manner."

Guidelines for Responsible Use:

1

Set time limits

Don't spend excessive time on devices

2

Protect personal information

Be careful what you share online

3

Build safe friendships

Be cautious about online relationships

4

Use antivirus software

Protect your devices

5

Visit appropriate websites

Avoid harmful content

6

Maintain privacy

Respect others' privacy

7

Balance online and offline life

Stay connected with real world

Discussion Point:

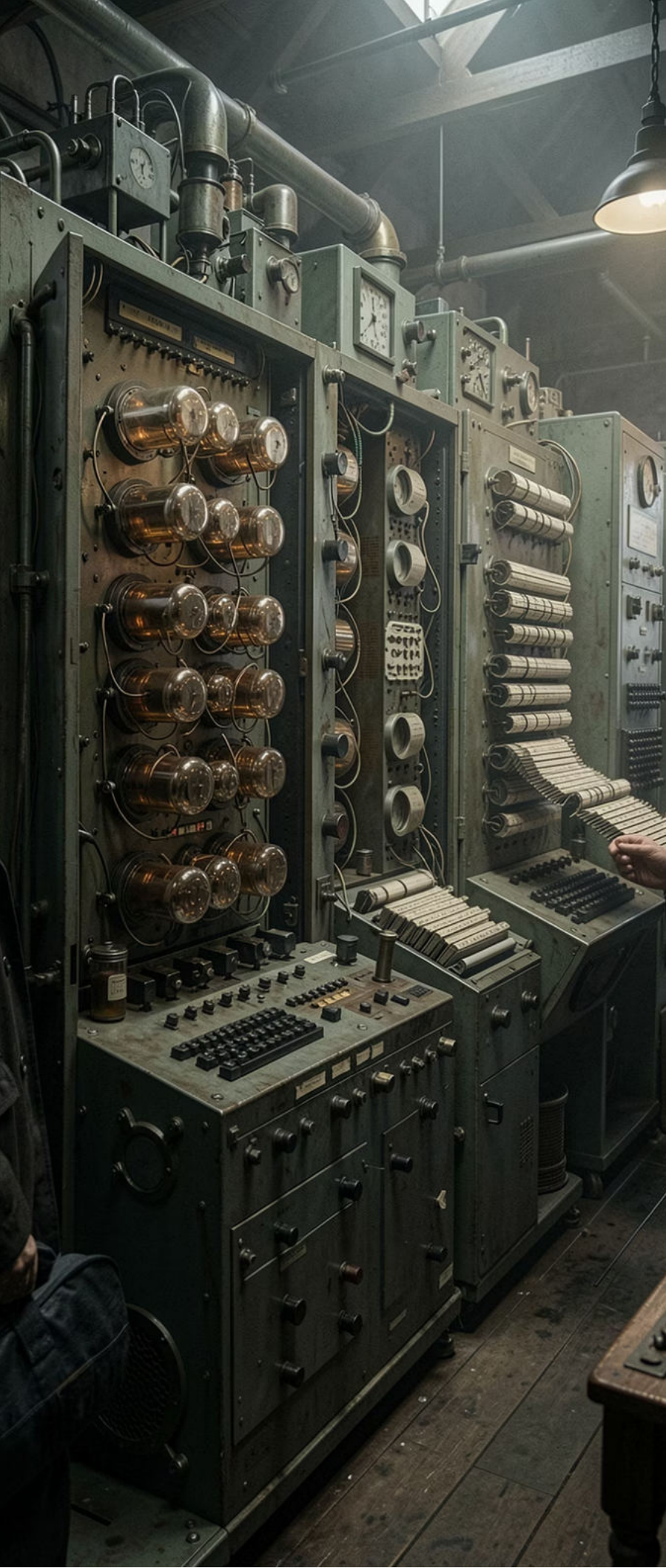
"What are some ways you practice responsible ICT use?"

Evolution of Computer - History

The computer was born with the attempt to make an adding machine.

Key Inventions Timeline:

Year	Invention	Inventor/Contributor	Significance
~5000 years ago	Abacus	Unknown	First calculating device
1642	Adding Machine	Blaise Pascal	World's first mathematical machine
1674	Improved Calculator	Leibnitz	Could perform multiplication and division
~1800s	Mechanical Loom (Punch Card System)	Joseph Jacquard	Used punch cards for instructions
1800s	Analytical Engine	Charles Babbage	Based on input-process-output-store concept; Father of Computing
1800s	First Programmer	Ada Augusta Lovelace	Wrote programs for Analytical Engine
1944	MARK 1 (Automatic Sequence Control Calculator)	Howard Aiken with IBM	Harvard University



Computer Generations - First Generation

Period: 1940-1956

Major Hardware:

Vacuum Tubes, Punch Cards

Software:

Machine Language, Assembly Language, Stored Program Concept

Characteristics:

- ✗ High Heat Generation
- ✗ Slow in processing
- ✗ Large in size
- ✗ Not Portable
- ✗ Consumes a lot of electricity
- ✗ Expensive

Key Computers:

- ENIAC
- EDVAC
- EDSAC
- UNIVAC
- IBM 701

Computer Generations - Second Generation

Period: 1956-1963

Major Hardware: Transistors, Tape, Floppy Disk (Secondary Storage)

Software: High-level Programming Language, Assembly Language

Characteristics:

-  Smaller in size
-  Low power consumption
-  Still expensive
-  Less heat generation
-  Faster processing

Key Computers:

- Honeywell 400
- IBM 7030
- CDC 1604
- UNIVAC LARC

Key Improvement:

"Transistors replaced vacuum tubes, making computers smaller, faster, and more reliable"

Computer Generations - Summary Comparison

Feature	First Generation (1940-1956)	Second Generation (1956-1963)
Hardware	Vacuum Tubes	Transistors
Size	Very Large	Smaller
Heat	High Heat Generation	Less Heat
Speed	Slow	Faster
Electricity	High Consumption	Low Consumption
Cost	Very Expensive	Still Expensive but lower
Storage	Punch Cards	Tape, Floppy Disks
Software	Machine Language, Assembly Language	High-level Languages, Assembly Language

Note: "Further generations (3rd-5th) are typically studied in higher grades"

Emerging ICT Trends

Trends to Watch:

Artificial Intelligence (AI)	Machines learning and making decisions
Internet of Things (IoT)	Connected devices communicating
Cloud Computing	Data storage and services over internet
5G Technology	Faster mobile internet
Virtual Reality (VR)	Immersive digital experiences
Augmented Reality (AR)	Digital overlays on real world
Big Data	Analyzing large datasets
Cybersecurity	Protecting digital information

Connection to Textbook:

- Many applications discussed (telemedicine, robotics, smart farming) are part of these trends
- The evolution from 1st gen to current gen shows continuous improvement

Future of ICT

Predictions for Future:



Even more connected world



Advanced healthcare technology



Personalized education



More automation and AI



Sustainable and green technology



Global collaboration

Important Messages:

"It is our responsibility to use ICT in a responsible manner"

"Understanding ICT helps us adapt to future changes"



Module 1.3 Summary

Key Points from Module 1.3:

Topic	Key Takeaways
Benefits	Speed, reach, cost reduction, efficiency, access to information
Demerits	Addiction, health issues, viruses, privacy concerns
Responsible Use	Set limits, protect privacy, safe practices
Computer Evolution	Abacus → Pascal → Babbage → MARK 1
First Generation	Vacuum tubes, large, slow, expensive
Second Generation	Transistors, smaller, faster, less heat
Emerging Trends	AI, IoT, Cloud, 5G, VR/AR, Big Data

Review Questions:

1. List 3 benefits and 3 demerits of ICT
2. What are 3 guidelines for responsible ICT use?
3. Who is called the Father of Computing?
4. What was the main hardware in first generation computers?
5. How did second generation computers improve?

Final Activities & Discussion

Group Activity:

Find information on computer history and make a report as a group activity

Individual Activity:

1. List ICT applications you use daily
2. Categorize them as benefits or challenges
3. Write 3 rules for responsible ICT use

Discussion Questions:

- How has ICT changed your grandparents' vs your generation's life?
- What emerging ICT trend interests you most?
- How can we reduce negative impacts of ICT?

Reflection:

"As you move forward in your ICT journey, remember that technology is a tool. How we use it determines whether it brings positive or negative impacts to our lives and society."